

IADS Agentic AI Hackathon

Internal Topic Briefing for Organising Team

The hackathon will challenge students to build **AI-powered agents deployed on Oracle Cloud Infrastructure (OCI)** that address real-world enterprise problems. Teams will be given one of the predefined challenges and will have **48 hours to design, build, and deploy a working prototype agent**.

The solutions are expected to demonstrate the following capabilities:

- Retrieval-Augmented Generation (RAG)
- Integration with OCI AI services
- Use of vector databases for semantic search
- Deployment of an AI-enabled service or application

Students may use **OCI Generative AI services, OCI Data Science notebooks, OCI Object Storage, Autonomous Database, and API services** where relevant.

Topic 1: Contract Review Agent: Legal Document Summarisation and Risk Flagging

Aim: To build an AI-powered agent capable of analysing contracts or legal documents, generating concise summaries, and identifying potentially risky clauses. The system should assist users in quickly understanding the key elements of a contract and highlight sections that may require further review.

Objective - The agent should:

- Analyse long legal or contractual documents
- Generate clear summaries of key terms
- Identify clauses related to risk or obligations
- Highlight potential issues such as liability, termination conditions, or penalties
- Provide a structured output for users reviewing the contract

The goal is **not to replace legal expertise**, but to provide an **AI-assisted first-pass review tool**.

Suggested Functional Capabilities

Students may implement features such as:

- Contract summarisation
- Clause classification (e.g. payment terms, liability, termination)
- Risk keyword detection

- Question answering over contracts
- Highlighting important sections within documents

Advanced teams may implement:

- A conversational contract assistant
- Clause comparison between multiple contracts
- Risk scoring of documents

Example User Scenario: A small business uploads a supplier agreement. The AI agent:

1. Summarises the contract in plain English
2. Identifies key clauses (termination, liability, payment)
3. Flags potential risks such as:
 - unlimited liability
 - restrictive termination conditions
 - high penalties

Suggested Open Datasets: Students may build a vector database of legal documents using publicly available datasets.

Contract datasets

- Contract Understanding Atticus Dataset (CUAD)
- SEC EDGAR contracts
- Public procurement agreements

Legal text datasets

- EU legislation corpus
- UK legislation datasets
- Public legal contracts from Kaggle

These documents can be used to create embeddings and support semantic retrieval for clause detection or question answering.

Example System Architecture

Typical pipeline students may implement:

1. Document ingestion
2. Text chunking
3. Embedding generation
4. Vector database storage
5. Retrieval-Augmented Generation
6. LLM-based summarisation and analysis

7. Web interface for interaction

Topic 2: Customer Support Triage Agent: Intelligent Routing and Prioritisation of Customer Support Requests

Aim: To develop an AI-powered triage agent that automatically analyses incoming customer support messages, determines the nature and urgency of the request, and routes it to the most appropriate support team. The agent should help organisations reduce response time, improve service efficiency, and ensure that urgent or complex cases are prioritised correctly.

Specific Use Case: A growing SaaS or digital service company receives hundreds of customer requests daily through email, chat, and web forms. These requests are manually reviewed by support staff before being assigned to relevant teams such as billing, technical support, or account management. Manual triage creates several challenges:

- Delays in responding to urgent issues
- Misclassification of support tickets
- High workload for support teams
- Inconsistent prioritisation of customer issues

The proposed AI agent should automatically analyse incoming customer messages and perform an initial triage step before they reach human agents.

Objective: The agent should perform the following tasks:

1. **Classify the request type:** Identify the category of the request such as:
 - Billing or payment issues
 - Technical problems or bugs
 - Account access issues
 - Product enquiries
 - Complaints or service dissatisfaction
2. **Assess urgency and priority:** Detect signals that indicate urgency, such as:
 - Service outage
 - Failed payments
 - Security concerns
 - Highly negative sentiment
3. **Generate a short ticket summary:** Produce a concise summary of the issue to assist the support team.
4. **Route the ticket to the appropriate department**
5. *(Optional)* Suggest a **first-response message** or provide links to relevant help articles.

Example Scenario

A customer submits the following request: “I was charged twice for my subscription and now my account is locked. I need this fixed urgently.”

The AI agent should:

1. Detect the category: **Billing / Account Issue**
2. Identify urgency due to billing error and account access problem
3. Generate a short summary:
Customer reports duplicate payment and account lockout after subscription renewal.
4. Assign the ticket to the **Billing Support Team** with **high priority**.

Suggested Open Datasets: Students may build their solution using open datasets for intent classification, sentiment detection, and support ticket analysis. Examples include:

- **Customer Support Dataset:** Contains real-world customer support conversations between companies and customers.
- **Banking77 Dataset:** A dataset containing labelled customer service queries for intent classification.
- **Customer Complaint Database:** Public complaint datasets that include structured descriptions of customer issues.
- **Helpdesk Ticket Datasets:** Various publicly available ticket datasets from Kaggle that include issue descriptions and categories.

These datasets can be used to:

- Train or fine-tune classification models
- Build embeddings for semantic search
- Create vector databases for retrieval-based responses

Possible AI Capabilities: Students may implement one or more of the following:

- Intent classification
- Sentiment analysis
- Priority prediction
- Ticket summarisation
- Retrieval of relevant help articles (RAG system)

Example Architecture: A typical implementation may include:

1. Incoming support query (text input)
2. Preprocessing and embedding generation
3. Intent classification model
4. Sentiment analysis and urgency detection
5. Retrieval of relevant knowledge base documents

6. Ticket summary generation
7. Routing output and dashboard display

Expected Output: Teams should demonstrate:

- Automated classification of customer queries
- Prioritisation of requests
- Suggested routing to departments
- A simple user interface or dashboard showing the triage result

Topic 3: Natural Language Query Agent: AI-Powered Querying of Structured Database (can be applied to different data sets)

Aim: To build an AI-powered agent that enables users to query structured database using natural language instead of SQL. The system should translate user queries written in plain English into executable database queries and return accurate, structured results. This solution aims to make data access more intuitive and accessible to non-technical users.

Objective- The agent should:

- Accept natural language queries from users
- Generate accurate SQL queries
- Execute queries on structured databases (e.g. sales, customer, or inventory data)
- Return results in a structured table format
- Ability to summarize results based on user request

The goal is to enable faster and easier data access for business users through an AI-assisted interface.

Suggested Functional Capabilities

Students may implement features such as:

- Natural language to SQL translation.
- Validation and error handling.
- Interactive query refinement (using follow-up questions).
- Dashboard or UI for user interaction.

Advanced teams may implement:

- A conversational data assistant (multi-turn queries).
- Support for multiple databases or data sources.
- Query explanation by breaking down generated SQL for users.

- Optional: Create a Vector based RAG Solution to summarize specifications / documents associated with the dataset (e.g. technical specifications for products / Inventory, invoice or contract documents for financial dataset)

Example User Scenario:

A business user wants to analyze sales performance but does not know SQL or how the underlying data is stored.

User query: “What were the total sales in the UK region last quarter, broken down by product category?”

The AI agent should:

1. Interpret the intent (aggregation, filter, and grouping).
2. Map terms like “UK region”, “last quarter”, and “product category” to the database schema
3. Generate an SQL query internally.
4. Return results such as a table showing sales by product category.

Datasets or Data-

Students may use publicly available structured datasets from open sources (for example: Kaggle), or generate synthetic data tailored to the required attributes for their solution. The Example data types include:

- Retail Sales data
- Customer Data
- Inventory Data
- Financial Data
- E-Commerce Data

These datasets can be used to simulate realistic business scenarios and support query generation and testing.

Example System Architecture-

A typical pipeline students may implement:

1. User Input (Natural Language Query).
2. Schema Understanding (metadata and table relationships if using multiple table).
3. LLM-based SQL Query Generation (text to SQL).
4. Query Execution on Database.
5. Web Interface (UI) for User Interaction.